# Applied Graph Theory

MAT 4409

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# List of Symbols

| | |
|---|---|
| $V(G)$ | Vertex set of graph $G$ |
| $E(G)$ | Edge set of graph $G$ |
| $\sim$ | is adjacent to |
| $\deg(v)$ | Degree of vertex $v$ |
| $\delta(G)$ | Minimum degree of graph $G$ |
| $\Delta(G)$ | Maximum degree of graph $G$ |
| $\mathrm{ecc}(v)$ | Eccentricity of vertex $v$ |
| $\mathrm{rad}(G)$ | Radius of graph $G$ |
| $\mathrm{diam}(G)$ | Diameter of graph $G$ |
| $A(G)$ | Adjacency matrix of graph $G$ |
| $G - v$ | Graph obtained from $G$ by deleting vertex $v$ and all edges incident with $v$ |
| $G - e$ | Graph obtained from $G$ by deleting edge $e$ |
| $G + uv$ | Graph obtained from $G$ by adding an edge between non-adjacent vertices $u$ and $v$ |
| $K_n$ | Complete graph of order $n$ (size $\binom{n}{2}$) |
| $P_n$ | Path graph of order $n$ (size $n-1$) |
| $C_n$ | Cycle graph of order $n$ (size $n$) |
| $Q_k$ | Hypercube graph of order $2^k$ |

# 1 Algebraic and Structural Properties

## 1.1 Graphs – Review of the Basic Theory

We first review basic graph theory. Throughout these notes, all graphs are finite, simple, and undirected unless stated otherwise.

A graph is an ordered pair $G = (V, E)$ where $V$ is a non-empty set called the vertex set, whose elements are the vertices of $G$, and $E$ is a set of unordered pairs of distinct vertices of $G$, whose elements are the edges of $G$. The order of $G$ is its number of vertices and the size of $G$ is its number of edges.

Two vertices $u$ and $v$ of $G$ are adjacent, denoted by $u \sim v$, if $uv \in E$ (note that $uv$ here denotes an unordered pair of vertices, and $uv = vu$). If there is no edge between $u$ and $v$, then they are nonadjacent, denoted by $u \nsim v$. The vertex $v$ and the edge $uv$ are said to be incident with each other. The degree of a vertex $v$ is the number of edges incident with it, or equivalently, the number of vertices adjacent with it, and is denoted by $\deg(v)$.

**Lemma 1.1** (Handshaking Lemma)**.** *The sum of the degrees of all vertices of a graph is twice its size.*

*Proof.* Let $G$ be a graph. As each edge is incident with exactly two vertices, each edge is counted once by two terms in the sum of the vertex degrees. Thus,

$$\sum_{v \in V(G)} \deg v = 2|E(G)|. \qquad \square$$

**Corollary 1.2.** *The number of vertices of odd degree in a graph is always even.*

*Proof.* Exercise. $\square$

The minimum degree of $G$ is $\delta(G) = \min_{v \in V(G)} \deg v$, and the maximum degree of $G$ is $\Delta(G) = \max_{v \in V(G)} \deg v$. A regular graph is a graph

in which all vertices have the same degree. This common value of the degree is the regularity of the graph. A regular graph of regularity $k$ is a $k$-regular graph. A cubic graph is a 3-regular graph.

**Exercise 1.1.** Prove the following:

1. Any cubic graph has even order. For which other regular graphs will this statement hold?
2. In any graph on two or more vertices, at least two vertices have the same degree.
3. If $G$ is a graph of order $n$ and size $m$, then $\delta(G) \leq \frac{2m}{n} \leq \Delta(G)$.

## 1.2 Cartesian Products

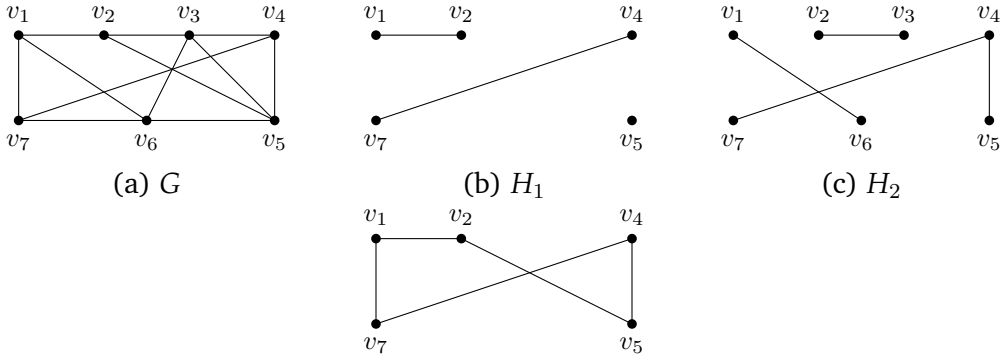
The Cartesian product of two graphs $G$ and $H$ is the graph denoted by $G \times H$, whose vertex set is $V(G) \times V(H)$ (the Cartesian product of the vertex sets $V(G)$ and $V(H)$, consisting of all ordered pairs $(u, v)$ where $u$ is a vertex of $G$ and $v$ is a vertex of $H$), in which two vertices $(u_1, v_1)$ and $(u_2, v_2)$ are adjacent if and only if either $u_1 = u_2$ and $v_1 \sim v_2$ or $u_1 \sim u_2$ and $v_1 = v_2$.

**Exercise 1.2.** Prove that if $G$ is a graph of order $p_1$ and size $q_1$, and $H$ is a graph of order $p_2$ and size $q_2$, then $G \times H$ is a graph of order $p_1 p_2$ and size $p_1 q_2 + p_2 q_1$.

## 1.3 Subgraphs

A subgraph of a graph $G$ is a graph $H$ whose vertex and edge sets are, respectively, subsets of the vertex and edge sets of $G$ – i.e. $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. A spanning subgraph of $G$ is a subgraph containing all the vertices of $G$. A subgraph $H$ of a graph $G$ is an induced subgraph if for all vertices $u, v \in V(H)$, $uv \in E(H)$ whenever $uv \in E(G)$ – i.e. every edge of $G$ that joins two vertices of $H$ is present in $H$.

**Example 1.3.** Fig. 1 shows a graph $G$ and three subgraphs $H_1$, $H_2$, and $H_3$ of $G$. The subgraph $H_1$ is neither a spanning subgraph (e.g. it does not contain $v_3$), nor an induced subgraph (e.g. it contains vertices $v_2$ and $v_5$ of $G$ but not the edge $v_2v_5$). $H_2$ is a spanning subgraph and $H_3$ is an induced subgraph of $G$.




(a) $G$ (b) $H_1$ (c) $H_2$

(d) $H_3$

Figure 1: A graph with three of its subgraphs

A graph is complete if all its vertices are adjacent to one another. The complete graph of order $n$ is denoted by $K_n$. As there is an edge between every pair of distinct vertices of $K_n$, the size of $K_n$ is $\binom{n}{2}$. A totally disconnected graph has no edges.

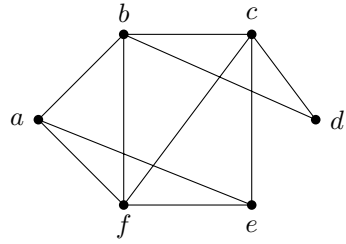
## 1.4 Walks, Paths, and Cycles

A walk in a simple graph $G$ is a sequence of vertices $v_1, v_2, \ldots, v_k$ such that $v_i \sim v_{i+1}$ for $i = 1, 2, \ldots, k-1$. This is a walk from $v_1$ to $v_k$ or between $v_1$ and $v_k$, having length $k-1$. We may also say that this is a walk of length $k-1$ starting at $v_1$ and ending at $v_k$.

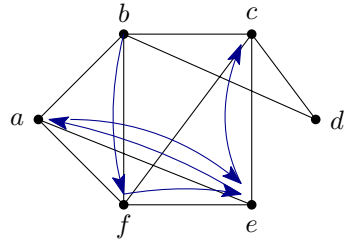
A walk in which no vertex is repeated is a path. Note that the length of a path is the number of edges in it.

A closed walk in $G$ is a walk starting and ending at the same vertex. A closed walk in which no vertex is repeated (except for the start and end vertices being the same) is a cycle. The length of a cycle is also equal to the number of vertices in it. A cycle of the form $v_1, \ldots, v_{k-1}, v_k$ is usually referred to as "the cycle $v_1, \ldots, v_{k-1}$".

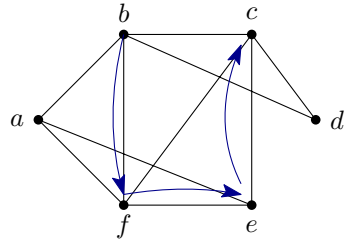
**Example 1.4.** Consider the graph $G$ shown below.



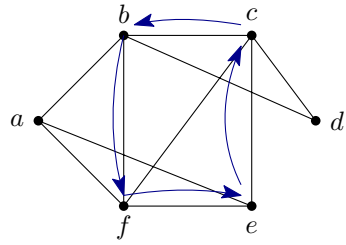

A walk of length $5$ from $b$ to $c$ in $G$ is $b, f, e, a, e, c$.




This walk is not a path, since the vertex $e$ is repeated. An example of a path from $b$ to $c$ in the same graph $G$ is $b, f, e, c$. The length of this path is $3$.



A cycle of length 4 in $G$ is $b, f, e, c, b$.




**Theorem 1.5.** *If $\delta(G) \geq 2$, then $G$ contains a path of length $\delta(G)$ and a cycle of length greater than $\delta(G)$.*

*Proof.* Let $P = v_0 v_1 \cdots v_k$ be a longest path in $G$. Then all the neighbours of $v_0$ must be on $P$ (for otherwise, we would get a path longer than $P$). Hence, $k \geq \deg v_0 \geq \delta(G)$, so $P$ has length at least $\delta(G)$.

Next, let $i \leq k$ be maximal such that $v_i \sim v_0$. Since all $\deg v_0$ neighbours of $v_0$ occur among $v_1, \ldots, v_i$, we have $i \geq \deg v_0 \geq \delta(G)$. Therefore, $v_0 v_1 \cdots v_i v_0$ is a cycle of length $i + 1 > \delta(G)$. □

**Exercise 1.3.** Prove that any two longest paths in a connected graph share at least one common vertex.

## 1.5 Distances and Connectedness

The distance between two vertices $u$ and $v$ of a graph $G$ is the length of a shortest path between $u$ and $v$ – such a shortest path between $u$ and $v$ is a geodesic from $u$ to $v$. Thus, the distance between $u$ and $v$ in $G$, denoted by $d_G(u, v)$, is the length of a geodesic from $u$ to $v$ in $G$. If there is no path from $u$ to $v$ in $G$, then we set $d_G(u, v) = \infty$. When the graph $G$ is clear from the context, we will write $d_G(u, v)$ as $d(u, v)$.

A graph is connected if there is a path between every two of its vertices. Otherwise, it is disconnected. A (connected) component of a graph

$G$ is a maximal connected subgraph of $G$. Thus, a graph $G$ is connected if and only if it has exactly one component.

The eccentricity of a vertex $v$, denoted by $\mathrm{ecc}(v)$, is the maximum of all distances from $v$ to any other vertex. That is,

$$\mathrm{ecc}(v) = \max_{u \in V(G)} d(u, v).$$

The diameter of $G$ is the maximum of the eccentricities of vertices of $G$, and the radius of $G$ is the minimum of the eccentricities of vertices of $G$. They are denoted by $\mathrm{diam}(G)$ and $\mathrm{rad}(G)$ respectively. Thus,

$$\mathrm{diam}(G) = \max_{v \in V(G)} \mathrm{ecc}(v) = \max_{u,v \in V(G)} d(u, v)$$
$$\mathrm{rad}(G) = \min_{v \in V(G)} \mathrm{ecc}(v) = \min_{v \in V(G)} \max_{u \in V(G)} d(u, v).$$

The set of all vertices of $G$ having minimum eccentricity,i.e. the set of all $v \in V(G)$ such that $\mathrm{ecc}\, v = \mathrm{rad}\, G$, is the centre of $G$.

**Exercise 1.4.** Prove that for any connected graph $G$, $\mathrm{rad}\, G \leq \mathrm{diam}\, G \leq 2\,\mathrm{rad}\, G$.

## 1.6 Bipartite Graphs

A graph $G = (V, E)$ is bipartite if its vertex set $V$ can be partitioned into two (possibly empty) subsets $V_1$ and $V_2$ such that no two vertices in $V_i$ are adjacent, for $i = 1, 2$. That is, there exist two disjoint subsets $V_1, V_2 \subseteq V$ such that $V_1 \cup V_2 = V$, and every edge of $G$ (if any) joins a vertex in $V_1$ with a vertex in $V_2$. Then we say that $(V_1, V_2)$ is a bipartition of $G$.

The following theorem characterises bipartite graphs in terms of their cycles. A cycle is even if its length is even and odd if its length is odd.

**Theorem 1.6.** *A graph is bipartite if and only if it contains no odd cycles.*

The complete bipartite graph $K_{m,n}$ is the bipartite graph with $m$ vertices in one partite set and $n$ vertices in the other partite set, such that every vertex of one partite set is adjacent to every vertex of the other.

## 1.7 Trees

A tree is a connected, acyclic graph. There are several well-known characterisations or alternative definitions of trees. We take the given definition as the basic one and prove its equivalence to some others. If $u$ and $v$ are non-adjacent vertices of a graph $T$, then $T + uv$ denotes the graph obtained from $T$ by adding the edge $uv$.

**Theorem 1.7.** *If $T$ is a graph of order $p$ and size $q$, the following are equivalent:*

*(i)* *$T$ is a tree.*
*(ii)* *Any two vertices of $T$ are joined by a unique path.*
*(iii)* *$T$ is connected and $p = q + 1$.*
*(iv)* *$T$ is acyclic and $p = q + 1$.*
*(v)* *$T$ is connected and for any edge $e$ of $T$, $T - e$ is disconnected.*
*(vi)* *$T$ is acyclic and for any two non-adjacent vertices $u$ and $v$ of $T$, $T+uv$ contains a unique cycle.*
*(vii)* *$T$ is connected and either of order at most 2, or not complete, and for any two non-adjacent vertices $u$ and $v$ of $T$, $T+uv$ contains a unique cycle.*

**Exercise 1.5.** A pendant vertex of a graph is a vertex of degree 1. Prove that every non-trivial tree contains at least two pendant vertices.

**Hint:** If a tree of order $p \geq 2$ had at most one vertex of degree 1, its degree sum would be at least $1 + 2(p-1) = 2p - 1$, contradicting the Handshaking Lemma and the fact that its size is $p - 1$.

**Exercise 1.6.** The centre of a graph $G$ is the set of all vertices of $G$ with minimum eccentricity – i.e. the set of all vertices $v$ of $G$ with $\operatorname{ecc} v = \operatorname{rad} G$. Show that every tree has a centre consisting of either exactly one

vertex or exactly two adjacent vertices.

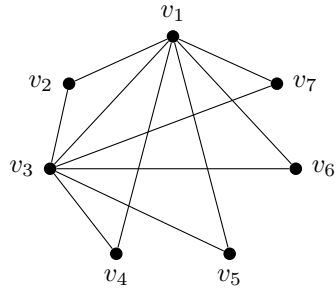
**Hint:** Handle trees of orders 1 and 2 separately. Deleting all pendant vertices from a tree of order at least 3 results in a smaller tree with the same centre.

## 1.8 Adjacency Matrices

The adjacency matrix of a graph $G$ of order $n$, with vertex set $V = \{v_1, \ldots, v_n\}$, is the $n \times n$ matrix $A = A(G)$ whose $(i, j)$-entry is

$$a_{ij} = \begin{cases} 1, & v_i \sim v_j \\ 0, & v_i \nsim v_j. \end{cases}$$

**Example 1.8.** The adjacency matrix of the graph




is

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Observe that, as the graphs we discuss are simple graphs and therefore have no self-loops on vertices, no vertex is adjacent to itself –i.e.

$a_{ii} = 0$ for all $i = 1, \ldots, n$. Also, since the graphs are undirected, $v_i \sim v_j$ if and only if $v_j \sim v_i$ – i.e. $a_{ij} = a_{ji}$. Thus, we have the following observation.

**Observation 1.9.** *The adjacency matrix of a (simple, undirected) graph is a symmetric, zero-diagonal,* $(0, 1)$*-matrix.*

In the $i^{\text{th}}$ row of the adjacency matrix, for each $j$, the $j^{\text{th}}$ entry is $1$ if $v_j$ is adjacent to $v_i$, and $0$ otherwise. That is, the number of $1$s in the $i^{\text{th}}$ row is the number of vertices adjacent to $v_i$, or in other words, the degree of $v_i$. Thus, the row sums of $A$ are the vertex degrees. Observe that $A\mathbb{1}$ is the vector of row sums, where $\mathbb{1}$ is the vector (of suitable size) with all entries equal to $1$. For instance, with the matrix $A$ given in Example 1.8,

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 6 \\ 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}.$$

From the Handshaking Lemma and the preceding observation, it follows that the sum of all entries of $A$ is twice the number of edges of the graph.

**Theorem 1.10.** *Let $A$ be the adjacency matrix of a graph $G$ with vertex set $\{v_1, \ldots, v_n\}$. Then the $(i, j)$-entry of $A^m$, for any positive integer $m$, is the number of walks of length $m$ from $v_i$ to $v_j$.*

*Proof.* The $(i, j)$-entry of $A^m$ is

$$(A^m)_{ij} = \sum_{1 \le k_1, \ldots, k_{m-1} \le n} a_{ik_1} a_{k_1 k_2} \cdots a_{k_{m-1} j}$$

Each term in this sum is $1$ if $a_{ik_1} = \cdots = a_{k_{m-1} j} = 1$ – i.e. if $v_i \sim v_{k_1} \sim \cdots \sim v_{k_{m-1}} \sim v_j$ – and is $0$ otherwise. Thus, each term counts a walk of length $m$ from $v_i$ to $v_j$. □

**Theorem 1.11.** *Let $A$ be the adjacency matrix of a graph $G$ with vertex set $\{v_1, \ldots, v_n\}$. Then*

*(i)* $\operatorname{tr}(A^2) = 2|E(G)|$
*(ii)* $\operatorname{tr}(A^3) = 6c_3(G)$

*where $c_3(G)$ denotes the number of triangles in $G$.*

*Proof.* We know that $(A^m)_{ij}$ is the number of walks of length $m$ from $v_i$ to $v_j$.

(i) Hence, the $i^{\text{th}}$ diagonal entry of $A^2$, viz. $(A^2)_{ii}$, is the number of walks of length $2$ from $v_i$ to itself. Any walk of length $2$ from $v_i$ to itself is of the form $v_iv_jv_i$, where $v_j$ is a vertex adjacent to $v_i$. Thus, the number of such walks is equal to the number of vertices adjacent to $v_i$,i.e. $\deg v_i$. Hence, $\operatorname{tr}(A^2) = \sum_{i=1}^{n} \deg v_i = 2|E(G)|$, by Handshaking Lemma.

(ii) Similarly, $(A^3)_{ii}$ is the number of walks of length $3$ from $v_i$ to itself. Any such walk is of the form $v_iv_jv_kv_i$, which implies that $v_i$, $v_j$, and $v_k$ form a triangle in $G$. Moreover, each such triangle corresponds to two distinct walks from $v_i$ to itself, when traversed in the two opposite directions. Thus, $(A^3)_{ii}$ is twice the number of triangles having $v_i$ as one of its vertices. Therefore, $\operatorname{tr}(A^3)$ is $6c_3(G)$, since each triangle contains three vertices, each of which counts the triangle twice in the sum $\operatorname{tr}(A^3)$. □

## 1.9 Graph Spectra

The (adjacency) spectrum of a graph is the spectrum (i.e. the multiset of eigenvalues) of its adjacency matrix. If the distinct eigenvalues of $A(G)$ are $\lambda_1, \ldots, \lambda_k$, with respective multiplicities $m_1, \ldots, m_k$, then we write

$$\operatorname{spec} G = \begin{pmatrix} \lambda_1 & \lambda_2 & \cdots & \lambda_k \\ m_1 & m_2 & \cdots & m_k \end{pmatrix}.$$

**Exercise 1.7.** Show that a graph $G$ of order $n$ is $k$-regular graph if and only if the all-ones vector $\mathbb{1}_n$ is an eigenvector of $A(G)$ corresponding to the eigenvalue $k$.

### Spectrum of the Complete Graph

The adjacency matrix of $K_n$ is $J_n - I_n$, where $J_n$ is the $n \times n$ all-ones matrix. Since the eigenvalues of $J_n$ are $n$ with multiplicity $1$ and $0$ with multiplicity $n-1$, the spectrum of $K_n$ is

$$\operatorname{spec} K_n = \begin{pmatrix} n-1 & -1 \\ 1 & n-1 \end{pmatrix}.$$

**Exercise 1.8.** Verify that $\mathbb{1}_n$ is an eigenvector of $A(K_n)$ corresponding to the eigenvalue $n-1$, and the column vectors $\begin{bmatrix} 1 & 0 & \cdots & -1 & \cdots & 0 \end{bmatrix}^T$, where the $-1$ occurs in the $i^{\text{th}}$ position, $i = 2, \ldots, n$, are independent eigenvectors corresponding to the eigenvalue $-1$.

### Spectrum of the Complete Bipartite Graph

The spectrum of the complete bipartite graph $K_{m,n}$ is

$$\operatorname{spec} K_{m,n} = \begin{pmatrix} \sqrt{mn} & 0 & -\sqrt{mn} \\ 1 & m+n-2 & 1 \end{pmatrix}.$$

### Spectrum of the Cycle

The eigenvalues of the cycle $C_n$, $n \geq 3$, are

$$\lambda_k = 2\cos\left(\frac{2k\pi}{n}\right),\ k = 0, \ldots, n-1.$$

**Spectrum of the Path**

The eigenvalues of the path $P_n$ are

$$\lambda_k = 2\cos\left(\frac{k\pi}{n+1}\right),\ k = 1, \ldots, n.$$

## 1.10 Application: Centrality and Clustering

A social network can be modelled by a graph whose vertices represent people and whose edges represent relationships or interactions. Different numerical measures capture different meanings of "importance", so identifying an influencer depends on the purpose of the analysis.

### 1.10.1 Centrality

Let $G$ be a connected graph of order $n \geq 2$. The following are common measures of the centrality of a vertex $v$.

(i) The degree centrality

$$C_D(v) = \frac{\deg(v)}{n-1}$$

measures the proportion of the network directly connected to $v$. A large value suggests high immediate reach.

(ii) The closeness centrality

$$C_C(v) = \frac{n-1}{\displaystyle\sum_{\substack{u \in V(G) \\ u \neq v}} d(u,v)}$$

is large when $v$ is only a few steps away from the rest of the network. Such a vertex can spread information efficiently.

(iii) For distinct vertices $s, t$, let $\sigma_{st}$ be the number of shortest $s$–$t$ paths and let $\sigma_{st}(v)$ be the number of these paths that pass through $v$. The betweenness centrality

$$C_B(v) = \sum_{\{s,t\} \subseteq V(G) - \{v\}} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

measures how often $v$ acts as a bridge between other vertices. It is particularly useful for finding brokers or potential bottlenecks.

(iv) An eigenvector centrality vector is a non-negative vector $\mathbf{x}$ satisfying

$$A\mathbf{x} = \lambda \mathbf{x},$$

where $A$ is the adjacency matrix and $\lambda$ is its largest eigenvalue. Its entry $x_v$ assigns a high score to $v$ when $v$ is adjacent to other high-scoring vertices.

*Remark.* No single centrality measure always identifies the "most influential" vertex. Degree measures direct popularity, closeness measures rapid access, betweenness measures brokerage, and eigenvector centrality values connections to already important vertices.

### 1.10.2 Clustering (Optional)

For a vertex $v$, let $N(v)$ be its set of neighbours and let $e(N(v))$ be the number of edges having both end-vertices in $N(v)$. The local clustering coefficient of $v$ is

$$C(v) = \begin{cases} \dfrac{e(N(v))}{\binom{\deg(v)}{2}}, & \deg(v) \geq 2, \\ 0, & \deg(v) < 2. \end{cases}$$

It is the proportion of pairs of neighbours of $v$ that are themselves adjacent. The average clustering coefficient of the network is

$$\overline{C}(G) = \frac{1}{n} \sum_{v \in V(G)} C(v).$$

A large local coefficient suggests that $v$ belongs to a tightly connected community. A vertex with high centrality but relatively low clustering may instead connect different communities.

**Exercise 1.9.** For the graph in Example 1.8, compute the degree and closeness centralities and the local clustering coefficient of every vertex. Which vertices would be selected as influencers by these measures?

**Exercise 1.10.** Explain how the entries of $A^2$ and $A^3$ can help detect common neighbours and triangles in a social network.

# 2 Network Reliability and Routing

A communication, transport, or supply network may be represented by a graph. Vertices represent terminals, routers, or locations, while edges represent links or routes. Removing a vertex models the failure of a terminal and all its incident links; removing an edge models the failure of one link. This section studies structural points of failure, economical connected backbones, and shortest-path routing.

## 2.1 Cutvertices, Cutedges, and Blocks

Let $c(G)$ denote the number of components of a graph $G$. For $v \in V(G)$, the graph $G - v$ is obtained by deleting $v$ and every edge incident with it. For $e \in E(G)$, the graph $G - e$ is obtained by deleting only $e$.

A vertex $v$ is a cutvertex if $c(G - v) > c(G)$. An edge $e$ is a cutedge, or a bridge, if $c(G - e) > c(G)$. Thus, if $G$ is connected, a cutvertex or cutedge is one whose deletion disconnects $G$.

**Theorem 2.1.** *An edge of a graph is a cutedge if and only if it lies on no cycle.*

*Proof.* Let $e = uv$. If $e$ lies on a cycle, the remaining part of that cycle is a $u$-$v$ path in $G - e$. Any path that uses $e$ can therefore be modified to avoid $e$, so deleting $e$ does not increase the number of components.

Conversely, suppose that $e$ lies on no cycle. If $u$ and $v$ were joined by a path in $G - e$, then that path together with $e$ would form a cycle containing $e$, which is not possible. Hence $u$ and $v$ lie in different components of $G - e$, and thus $e$ is a cutedge. □

**Theorem 2.2.** *If $G$ is connected and $v \in V(G)$, then the following are equivalent:*

*(i)* *$v$ is a cutvertex of $G$.*
*(ii)* *There is a partition of $V(G) - \{v\}$ into non-empty sets $U$ and $W$ such that every $u$-$w$ path passes through $v$ whenever $u \in U$ and $w \in W$.*
*(iii)* *There exist vertices $u, w \in V(G) - \{v\}$ such that every $u$-$w$ path passes through $v$.*

*Proof.* (i) $\implies$ (ii). Since $v$ is a cutvertex, the graph $G - v$ is disconnected, i.e. it has two or more components. Let $U$ be the set of all the vertices in any one of the components, and let $W$ be the set of all the remaining vertices of $G - v$. Clearly, $\{U, W\}$ is a partition of $V(G) - \{v\}$. Now, if $u \in U$ and $w \in W$, then $u$ and $w$ are in different components of $G - v$, which implies that any path from $u$ to $w$ must pass through $v$.
(ii) $\implies$ (iii) is obvious as the latter is a special case of the former.
(iii) $\implies$ (i). Consider the graph $G - v$. As every $u$-$w$ path passes through $v$, none of them is present in $G - v$, and therefore $G - v$ is disconnected. Hence, $v$ is a cutvertex of $G$. □

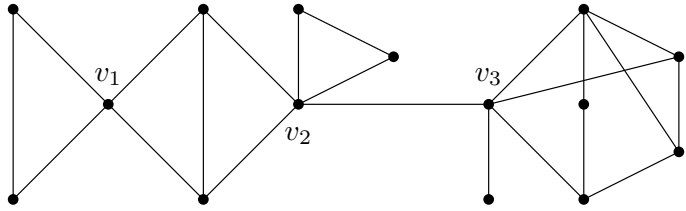
The following characterisation of bridges can be proved along similar lines to that of Theorem 2.2.

**Theorem 2.3.** *If $G$ is a connected graph, and $e$ is any edge of $G$, then the following are equivalent:*

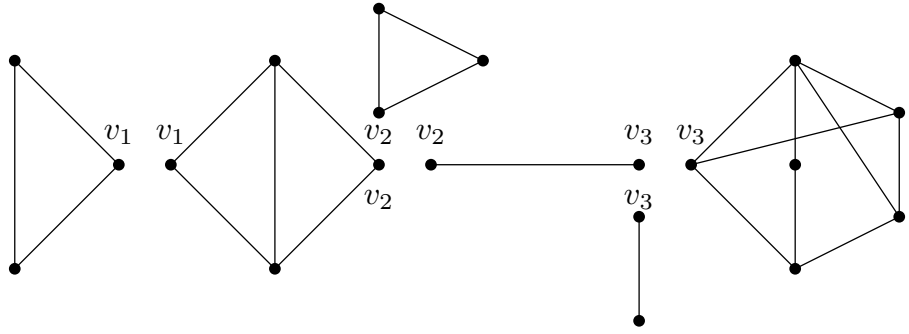
*(i)* *$e$ is a bridge of $G$.*
*(ii)* *$e$ is not on any cycle of $G$.*
*(iii)* *There exist vertices $u$ and $w$ of $G$ such that every $u$-$w$ path contains $e$.*
*(iv)* *There exists a partition of $V(G)$ into two non-empty subsets $U$ and $W$ such that for all $u \in U$ and $w \in W$, every $u$-$w$ path contains $e$.* □

A block of a graph is a maximal connected subgraph that has no cutvertex. With this convention, each cutedge together with its end-vertices is a block isomorphic to $K_2$, and each isolated vertex is a block isomorphic to $K_1$.

**Example 2.4.** The graph below has six blocks and three cutvertices, namely $v_1$, $v_2$, and $v_3$.




Its six blocks are shown separately below.



**Theorem 2.5.** *Let $G$ be a connected graph.*

*(i) Every edge of $G$ belongs to exactly one block.*
*(ii) Any two distinct blocks have at most one vertex in common.*
*(iii) A vertex belongs to at least two blocks if and only if it is a cutvertex.*

□

**Theorem 2.6.** *If $G$ is a connected graph of order at least 3, then the following are equivalent:*

*(i) $G$ is a block.*
*(ii) Every two vertices of $G$ lie on a common cycle.*
*(iii) Every vertex and edge of $G$ lie on a common cycle.*
*(iv) Every two edges of $G$ lie on a common cycle.* □

The block-cutvertex graph of a connected graph $G$ is the bipartite graph whose vertices are the blocks and cutvertices of $G$, with a block $B$ adjacent to a cutvertex $v$ precisely when $v \in V(B)$.

**Theorem 2.7.** *The block-cutvertex graph of a connected graph is a tree.* □

**Exercise 2.1.** Prove Theorem 2.5. Also prove that the block-cutvertex graph is a tree.

**Exercise 2.2.** Show that every edge of a tree is a cutedge, and that the cutvertices of a non-trivial tree are exactly its non-pendant vertices.

### 2.1.1 Application: Network Vulnerability Assessment

Cutvertices and cutedges represent single points of failure: the failure of one such component separates terminals that could previously communicate. Blocks describe regions with no single vulnerable vertex, while the block-cutvertex tree records how these regions are joined.

The number of resulting components is only one measure of the severity of a failure. Suppose that $G$ is connected and that the components of $G - v$ have orders $s_1, \ldots, s_k$. The number of unordered pairs of surviving vertices separated by the failure of $v$ is

$$P(v) = \sum_{1 \le i < j \le k} s_i s_j.$$

If a cutedge $e$ separates $G$ into components of orders $s$ and $n - s$, then the corresponding number of separated pairs is

$$P(e) = s(n - s).$$

These quantities distinguish, for example, a bridge that isolates one terminal from a bridge that divides the network into two large pieces.

*Remark.* This is a structural, deterministic model. A practical reliability assessment may also need failure probabilities, capacities, repair times, and the possibility of several simultaneous failures. Moreover, a vertex with high centrality need not be a cutvertex, and a cutvertex need not have high degree.

**Exercise 2.3.** For the graph in Example 2.4, determine $P(v)$ for each cutvertex and $P(e)$ for each cutedge. Rank these single points of failure from most to least disruptive.

## 2.2 Connectivity

For $S \subseteq V(G)$, the graph $G - S$ is obtained by deleting the vertices of $S$ and all edges incident with them. For $S \subseteq E(G)$, it is obtained by deleting the edges of $S$.

A set $S$ of vertices of a connected graph $G$ is a separating set or vertex cut of $G$ if $G - S$ is disconnected or trivial (i.e. a graph of order 1). The vertex connectivity (or connectivity) of $G$, denoted by $\kappa(G)$ (or $\kappa$), is the

cardinality of a minimum separating set of $G$ – i.e. the minimum number of vertices of $G$ whose removal results in a disconnected or trivial graph. Similarly, a disconnecting set of a connected graph $G$ is a set $S$ of edges of $G$ such that $G - S$ is disconnected or trivial. The cardinality of a minimum disconnecting set of $G$ is the edge connectivity of $G$, and is denoted by $\lambda(G)$ (or $\lambda$). The vertex connectivity and edge connectivity of a disconnected graph are defined to be $0$.

**Theorem 2.8.** *For any graph $G$, $\kappa(G) \leq \lambda(G) \leq \delta(G)$.*

*Proof.* If $\delta(G) = 0$, then clearly, $\lambda(G) = 0$. If $\delta(G) \geq 1$, then disconnecting all the edges incident with a vertex of degree $\delta(G)$ disconnects $G$, and hence $\lambda(G) \leq \delta(G)$.

If $G$ is disconnected, then $\kappa(G) = \lambda(G) = 0$. If $\lambda(G) = 1$, then $G$ has a bridge $e = uv$. If $G \cong K_2$, deleting either vertex leaves $K_1$. Otherwise, at least one of $u$ and $v$ is a cutvertex. Hence, in either case, $\kappa(G) = 1$.

Next, let $F$ be a minimum disconnecting set of $\lambda \geq 2$ edges, and choose $e = uv \in F$. By the minimality of $F$, the graph $H = G - (F - \{e\})$ is connected, while $H - e = G - F$ is disconnected. Hence, $e$ is a bridge of $H$.

For each edge of $F - \{e\}$, choose an end-vertex distinct from $u$ and $v$, and let $X$ be the set of chosen vertices. Then $|X| \leq \lambda - 1$, and all the edges of $F - \{e\}$ are absent from $G - X$. If $G - X$ is disconnected, then $X$ is a vertex cut. Otherwise, $e$ remains a bridge of $G - X$; deleting a suitable end-vertex of $e$ then leaves a disconnected or trivial graph. In either case, $G$ has a vertex cut of cardinality at most $\lambda$, and hence $\kappa \leq \lambda$. □

**Theorem 2.9.** *For all positive integers $a$, $b$, and $c$ such that $a \leq b \leq c$, there exists a graph $G$ with $\kappa(G) = a$, $\lambda(G) = b$, and $\delta(G) = c$.* □

It is easy to see that $\kappa(K_n) = n - 1$ and $\kappa(K_{p,q}) = \min(p, q)$. The $k$-dimensional hypercube $Q_k$ is the Cartesian product of $k$ copies of $K_2$, with $Q_0 = K_1$. The following theorem gives its connectivity.

**Theorem 2.10.** *For any $k \geq 0$, the connectivity of the hypercube $Q_k$ is $\kappa(Q_k) = k$.*

*Proof.* As $Q_k$ is $k$-regular, $\kappa(Q_k) \leq k$. We prove the reverse inequality by induction on $k$. The result is immediate for $k = 0$ and $k = 1$.

Let $k \geq 2$, suppose the result holds for $Q_{k-1}$, and write $Q_k$ as two copies $G_1$ and $G_2$ of $Q_{k-1}$ joined by a perfect matching. Suppose that $S \subseteq V(Q_k)$ and $|S| \leq k - 1$, and put $S_i = S \cap V(G_i)$ for $i = 1, 2$. We show that $Q_k - S$ is connected.

If $|S_1| \leq k - 2$ and $|S_2| \leq k - 2$, then both $G_1 - S_1$ and $G_2 - S_2$ are connected by the induction hypothesis. At least one edge of the perfect matching has neither end-vertex in $S$, so these two subgraphs are joined in $Q_k - S$.

Otherwise, say $|S_1| = k - 1$. Then $S_2 = \varnothing$, so $G_2$ is intact and connected. Every surviving vertex of $G_1$ has its matching neighbour in $G_2$, and hence $Q_k - S$ is connected. Thus, no set of at most $k - 1$ vertices is a vertex cut, so $\kappa(Q_k) \geq k$. □

**Corollary 2.11.** *For any $k \geq 0$, the edge connectivity of the hypercube $Q_k$ is $\lambda(Q_k) = k$.*

*Proof.* From Theorem 2.8, $k = \kappa(Q_k) \leq \lambda(Q_k) \leq \delta(Q_k) = k$. □

**Theorem 2.12.** *If $G$ is 3-regular, then $\kappa(G) = \lambda(G)$.*

*Proof.* If $G$ is disconnected, then $\kappa(G) = \lambda(G) = 0$. If $G \cong K_4$, then $\kappa(G) = \lambda(G) = 3$. We may therefore suppose that $G$ is connected and not complete.

Let $S$ be a minimum vertex cut of $G$. We know that $\lambda(G) \geq \kappa(G) = |S|$. The graph $G - S$ has at least two components, and the minimality of $S$ implies that every vertex of $S$ has a neighbour in every component of $G - S$.

If $G - S$ has three components, then every vertex of $S$ has exactly one neighbour in each component. The set of edges joining $S$ to any one component is therefore a disconnecting set of size $|S|$.

Now suppose that $G - S$ has two components, $H_1$ and $H_2$. A vertex of $S$ with a neighbour in $S$ has exactly one neighbour in each $H_i$; delete its edge to $H_1$. A vertex of $S$ with no neighbour in $S$ has one neighbour in one of the $H_i$ and two in the other; delete its edge to the component in which it has only one neighbour. Exactly one edge has now been deleted for each vertex of $S$, and no remaining path joins $H_1$ to $H_2$. Thus, $\lambda(G) \leq |S| = \kappa(G)$, and equality follows. □

## 2.3 Minimum Spanning Trees

A tree is a connected graph containing no cycles. A spanning tree of a connected graph $G$ is a spanning subgraph of $G$ that is a tree. If every edge $e$ of $G$ has a real weight $w(e)$, the weight of a subgraph $H$ is

$$w(H) = \sum_{e \in E(H)} w(e).$$

A minimum spanning tree of $G$ is a spanning tree having minimum weight among all spanning trees of $G$.

For a non-empty proper subset $S$ of $V(G)$, the edges with one end-vertex in $S$ and the other in $V(G) - S$ form an edge cut of $G$.

**Theorem 2.13** (Cut Property)**.** *Let $e$ be an edge of minimum weight in an edge cut of a connected weighted graph $G$. Then some minimum spanning tree contains $e$. If $e$ is the unique minimum-weight edge in the cut, every minimum spanning tree contains $e$.*

*Proof.* Let the cut be determined by $S \subset V(G)$, and let $T$ be a minimum spanning tree. There is nothing to prove if $e \in E(T)$. Otherwise, adding $e$ to $T$ creates a unique cycle. As $e$ crosses the cut, this cycle contains another edge $f$ that crosses the cut. Since $e$ has minimum weight in the cut, $w(e) \leq w(f)$. Replacing $f$ by $e$ therefore gives a spanning tree of weight at most $w(T)$, and hence another minimum spanning tree. If

$e$ is the unique minimum-weight edge in the cut, then $w(e) < w(f)$; consequently, a minimum spanning tree omitting $e$ would not be minimum. □

Prim's algorithm grows a tree from an arbitrary starting vertex. At each step, it chooses a cheapest edge joining a vertex already reached to a vertex not yet reached.

**Algorithm 1** Prim's algorithm

**Input:** Connected weighted graph $G$ and starting vertex $r$

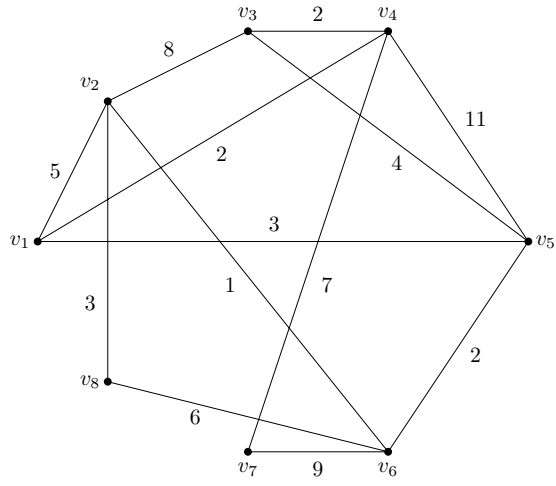
$S \leftarrow \{r\}$, $F \leftarrow \varnothing$
**while** $S \neq V(G)$ **do**
    Choose a minimum-weight edge $xy$ with $x \in S$ and $y \notin S$
    $F \leftarrow F \cup \{xy\}$
    $S \leftarrow S \cup \{y\}$
**end while**
**return** $T = (V(G), F)$

**Theorem 2.14.** *Prim's algorithm returns a minimum spanning tree.*

*Proof.* We maintain the invariant that some minimum spanning tree contains every edge in $F$. This is true initially. Suppose that a minimum spanning tree $T$ contains $F$, and let $e$ be the next edge chosen across the cut determined by $S$. If $e \in E(T)$, the invariant is preserved. Otherwise, adding $e$ to $T$ creates a unique cycle containing another edge $f$ that crosses the same cut. No edge of $F$ crosses this cut, and hence $f \notin F$. Since $e$ has minimum weight in the cut, $w(e) \leq w(f)$, so $T - f + e$ is a minimum spanning tree containing $F \cup \{e\}$. Thus, the invariant is preserved.

Each iteration adds one new vertex and one edge without creating a cycle. On termination, $F$ is itself a spanning tree; since some minimum spanning tree contains all its edges, $F$ is a minimum spanning tree. □

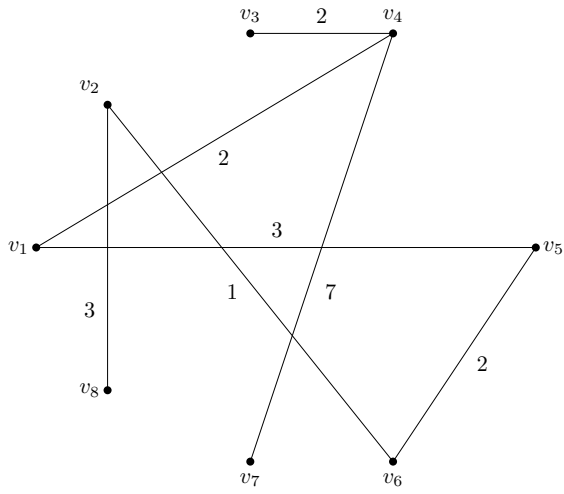
**Example 2.15.** Apply Prim's algorithm to the graph below, taking $v_1$ as the starting vertex.




One possible sequence of choices is shown in the table.

| Step | Edge added | Vertex added | Total weight |
|---|---|---|---|
| 1 | – | $v_1$ | 0 |
| 2 | $v_1v_4$ | $v_4$ | 2 |
| 3 | $v_4v_3$ | $v_3$ | 4 |
| 4 | $v_1v_5$ | $v_5$ | 7 |
| 5 | $v_5v_6$ | $v_6$ | 9 |
| 6 | $v_6v_2$ | $v_2$ | 10 |
| 7 | $v_2v_8$ | $v_8$ | 13 |
| 8 | $v_4v_7$ | $v_7$ | 20 |

The resulting minimum spanning tree has weight $20$.




*Remark.* A minimum spanning tree is a cheapest connected backbone, but it has no redundant edge: every edge of a tree is a cutedge. Cost minimisation and resilience are therefore different design objectives.

**Exercise 2.4.** Explain why Prim's algorithm can produce different minimum spanning trees when edge weights are repeated. Show that if all edge weights are distinct, the minimum spanning tree is unique.

## 2.4 All-Pairs Shortest Paths

Let $G$ be a weighted directed graph with vertices $v_1, \dots, v_n$. Its weighted adjacency matrix is $W = (w_{ij})$, where

$$w_{ij} = \begin{cases} 0, & i = j, \\ \text{weight of the edge from } v_i \text{ to } v_j, & (v_i, v_j) \in E(G), \\ \infty, & (v_i, v_j) \notin E(G). \end{cases}$$

We assume that $G$ has no negative-weight cycle. Edge weights may be negative. In the absence of a negative-weight cycle, a minimum-weight

walk can always be reduced to a path. If a negative-weight cycle is reachable from $v_i$ and can reach $v_j$, however, the weights of $v_i$–$v_j$ walks are unbounded below, so there is no finite shortest-path distance from $v_i$ to $v_j$.

For $k = 0, 1, \dots, n$, let $d_{ij}^{(k)}$ be the minimum weight of a $v_i$–$v_j$ path whose internal vertices belong to $\{v_1, \dots, v_k\}$. Initially, $D^{(0)} = W$. For $k \geq 1$, a shortest admissible path either avoids $v_k$, or passes through $v_k$. Therefore,

$$d_{ij}^{(k)} = \min\left\{ d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)} \right\}.$$

After the final stage, $d_{ij}^{(n)}$ is the distance from $v_i$ to $v_j$.

**Algorithm 2** Floyd–Warshall algorithm

**Input:** Weighted adjacency matrix $W$ of order $n$

```
D ← W
for k = 1, ..., n do
    for i = 1, ..., n do
        for j = 1, ..., n do
            d_ij ← min{d_ij, d_ik + d_kj}
        end for
    end for
end for
return D
```

**Theorem 2.16.** *If $G$ has no negative-weight cycle, the matrix returned by the Floyd–Warshall algorithm is its all-pairs distance matrix.*

*Proof.* After iteration $k$, the entry $d_{ij}$ equals $d_{ij}^{(k)}$. This follows by induction on $k$ from the recurrence above. At $k = n$, every vertex may be used internally, so each entry is the weight of a shortest path between the corresponding vertices. □

The algorithm performs $n^3$ comparisons and uses an $n \times n$ matrix. Hence, its running time is $O(n^3)$ and its storage requirement is $O(n^2)$. If some diagonal entry is negative after the algorithm terminates, the graph contains a negative-weight cycle.

## 2.5 Application: Routing Tables

A distance matrix gives route costs but not the routes themselves. To construct routing tables, maintain a next-hop matrix $N$. Initially, set $N_{ij} = j$ when $(v_i, v_j)$ is an edge, and leave $N_{ij}$ undefined when $i \neq j$ and there is no such edge. Whenever the Floyd–Warshall update through $v_k$ strictly improves $d_{ij}$, also set

$$N_{ij} \leftarrow N_{ik}.$$

Then $N_{ij}$ is the first vertex to visit on a shortest route from $v_i$ to $v_j$. Repeatedly consulting the appropriate row reconstructs the complete route.

**Example 2.17.** Consider the weighted adjacency matrix

$$W = \begin{bmatrix} 0 & 3 & \infty & 7 \\ 8 & 0 & 2 & \infty \\ 5 & \infty & 0 & 1 \\ 2 & \infty & \infty & 0 \end{bmatrix}.$$

Floyd–Warshall gives the distance and next-hop matrices

$$D = \begin{bmatrix} 0 & 3 & 5 & 6 \\ 5 & 0 & 2 & 3 \\ 3 & 6 & 0 & 1 \\ 2 & 5 & 7 & 0 \end{bmatrix}, \qquad N = \begin{bmatrix} - & 2 & 2 & 2 \\ 3 & - & 3 & 3 \\ 4 & 4 & - & 4 \\ 1 & 1 & 1 & - \end{bmatrix}.$$

For example, the entry $N_{14} = 2$ directs traffic from $v_1$ first to $v_2$. The successive next hops give the shortest route $v_1, v_2, v_3, v_4$, of total weight $6$.

**Exercise 2.5.** Apply Floyd–Warshall to the matrix in the preceding example, recording the matrix after each value of $k$. Verify the final distance and next-hop matrices.

**Exercise 2.6.** Suppose the weight of the edge from $v_3$ to $v_4$ in the preceding example increases from $1$ to $6$. Recompute the routing table and identify every source–destination pair whose shortest route changes.

# 3 Covering, Independence, and Matching

## 3.1 Coverings and Independence

A vertex $v$ and an edge $e$ of a graph $G$ cover each other if $v$ is incident with $e$. A vertex cover is a set of vertices that covers every edge. An edge cover is a set of edges that covers every vertex. Thus, a graph has an edge cover if and only if it has no isolated vertices.

A vertex (or edge) cover is minimal if no proper subset of itself is a vertex (or edge) cover. It is minimum if it has the least cardinality among all vertex (or edge) covers of the graph. The cardinality of a minimum vertex cover of $G$ is the vertex covering number of $G$, denoted by $\alpha_0(G)$, and the cardinality of a minimum edge cover of $G$ is the edge covering number of $G$, denoted by $\alpha_1(G)$.

A set of vertices or a set of edges of $G$ is independent if no two of its elements are adjacent. An independent set of vertices (or edges) is maximal if it is not properly contained in any larger independent set, and is maximum if it has the largest cardinality among all independent sets of vertices (or edges). The vertex independence number of $G$, denoted by $\beta_0(G)$, is the cardinality of a maximum independent set of vertices

of $G$. The edge independence number of $G$, denoted by $\beta_1(G)$, is the cardinality of a maximum independent set of edges of $G$.

**Example 3.1.** The values of the parameters $\alpha_0$, $\beta_0$, $\alpha_1$, and $\beta_1$ for some families of graphs are listed below.

| $G$ | $\alpha_0$ | $\beta_0$ | $\alpha_1$ | $\beta_1$ |
|---|---|---|---|---|
| $K_n$ | $n-1$ | $1$ | $\lceil \frac{n}{2} \rceil$ | $\lfloor \frac{n}{2} \rfloor$ |
| $K_{p,q}$ | $\min(p,q)$ | $\max(p,q)$ | $\max(p,q)$ | $\min(p,q)$ |
| $C_n$ | $\lceil \frac{n}{2} \rceil$ | $\lfloor \frac{n}{2} \rfloor$ | $\lceil \frac{n}{2} \rceil$ | $\lfloor \frac{n}{2} \rfloor$ |
| $P_n$ | $\lfloor \frac{n}{2} \rfloor$ | $\lceil \frac{n}{2} \rceil$ | $\lceil \frac{n}{2} \rceil$ | $\lfloor \frac{n}{2} \rfloor$ |
| $Q_k$ | $2^{k-1} = \frac{n}{2}$ | $2^{k-1} = \frac{n}{2}$ | $2^{k-1} = \frac{n}{2}$ | $2^{k-1} = \frac{n}{2}$ |

Here $n \geq 2$ for $K_n$ and $P_n$, $n \geq 3$ for $C_n$, $p, q \geq 1$, and $k \geq 1$ for $Q_k$.

**Exercise 3.1.** Justify the values of the parameters given in Example 3.1.

**Exercise 3.2.** Prove that a set $S$ of vertices of $G$ is independent if and only if $V(G) \setminus S$ is a vertex cover of $G$.

### 3.1.1 Gallai's theorem

The vertex covering and independence numbers are complementary. The same is true of the edge parameters whenever an edge cover exists.

**Theorem 3.2** (Gallai)**.** *For every graph $G$ of order $n$,*

$$\alpha_0(G) + \beta_0(G) = n.$$

*If $G$ has no isolated vertices, then*

$$\alpha_1(G) + \beta_1(G) = n.$$

*Proof.* Let $M_0$ be a maximum set of independent vertices of $G$. Then $\beta_0 = |M_0|$, and $V \setminus M_0$ is a vertex cover. Hence, $\alpha_0 \leq |V \setminus M_0| = n - \beta_0$. Let $N_0$ be a minimum cover of $G$. Then $\alpha_0 = |N_0|$, and $V \setminus N_0$ is independent. Hence, $\beta_0 \geq |V \setminus N_0| = n - \alpha_0$. Thus, $\alpha_0 + \beta_0 = n$.

Suppose now that $G$ has no isolated vertices, and let $M_1$ be a maximum set of independent edges. Since $M_1$ is maximal, no edge joins two vertices unsaturated by $M_1$. For each unsaturated vertex, choose an edge incident with it and add that edge to $M_1$. The chosen edges are distinct, so the resulting edge cover $Y$ has

$$|Y| = |M_1| + (n - 2|M_1|) = n - |M_1| = n - \beta_1.$$

Consequently, $\alpha_1 + \beta_1 \leq n$.

Next, let $N_1$ be a minimum edge cover, and let $H = (V(G), N_1)$. The graph $H$ is a forest: if it contained a cycle, deleting any edge of that cycle would leave an edge cover. Hence, $H$ has $n - |N_1|$ components. Taking one edge from each component gives an independent set of edges, and therefore

$$\beta_1 \geq n - |N_1| = n - \alpha_1.$$

Thus, $\alpha_1 + \beta_1 \geq n$, and equality follows. □

## 3.2 Matching

A matching of $G$ is a set of independent edges of $G$. Thus, the edge independence number $\beta_1(G)$ is also called the matching number of $G$. A vertex is saturated by a matching $M$ (it is $M$-saturated) if it is incident with some edge of $M$, and otherwise it is unsaturated by $M$ (it is $M$-unsaturated). A matching is

(i) maximal if it is not properly contained in any other matching.
(ii) maximum if it has greatest cardinality among all matchings.
(iii) perfect if it saturates all vertices.

A path of $G$ is $M$-alternating if its edges are alternately from $M$ and $E(G) \setminus M$. An $M$-augmenting path is an $M$-alternating path whose end-vertices are $M$-unsaturated.

*Note.* If $P$ is an $M$-augmenting path, we can obtain a matching of cardinality $|M| + 1$ by replacing the edges of $P$ in $M$ by the edges of $P$ not in $M$.

**Example 3.3.** Let $G$ be the bipartite graph with partite sets $\{a, b, c\}$ and $\{1, 2, 3\}$ and edge set

$$\{a1, a2, b1, c3\}.$$

For the matching $M = \{a1, c3\}$, the path $b, 1, a, 2$ is $M$-augmenting. Replacing $a1$ by $b1$ and $a2$ produces the perfect matching $\{b1, a2, c3\}$.

Recall that the symmetric difference of two sets $A$ and $B$ is $A \triangle B = (A \setminus B) \cup (B \setminus A)$. If $M$ and $M'$ are matchings of $G$, let $F$ be the edge-induced subgraph whose edge set is $M \triangle M'$ and whose vertex set consists of the end-vertices of these edges. We call $F$ the symmetric-difference graph of $M$ and $M'$.

**Lemma 3.4.** *Every component of the symmetric difference of two matchings is a path or an even cycle.*

*Proof.* Let $F$ be the symmetric-difference graph of $M$ and $M'$. As $M$ and $M'$ are matchings, every vertex has at most one incident edge from each of them. Hence, $\Delta(F) \leq 2$. The graph $F$ has no isolated vertices, so every component is a path or a cycle. Its edges alternate between $M \setminus M'$ and $M' \setminus M$, and hence every cycle has even length. □

**Theorem 3.5** (Berge)**.** *A matching $M$ of $G$ is maximum if and only if $G$ has no $M$-augmenting path.*

*Proof.* If $G$ has an $M$-augmenting path $P$, then the matching obtained from $M$ by removing the edges of $P$ that are in $M$ and adding those edges of $P$ that are not in $M$ has cardinality $|M| + 1$, which implies that

$M$ cannot be a maximum matching. Thus, if $M$ is a maximum matching, $G$ has no $M$-augmenting path.

Conversely, suppose that $M$ is not a maximum matching in $G$. Then $G$ has a matching $M'$ with $|M'| > |M|$. Let $F$ be their symmetric-difference graph. By Lemma 3.4, every component of $F$ is a path or an even cycle. Every cycle contains equally many edges from $M$ and $M'$. Since $|M'| > |M|$, some path component contains more edges from $M'$ than from $M$. Its first and last edges belong to $M'$, so its end-vertices are $M$-unsaturated. It is therefore an $M$-augmenting path. □

### 3.2.1 Hall's matching condition

Let $G$ be a bipartite graph with bipartition $(V_1, V_2)$. If $M$ is a matching of $G$ that saturates $V_1$ (i.e. saturates all vertices in $V_1$), then observe that for every subset $S \subseteq V_1$, $|N_G(S)| \geq |S|$, where $N_G(S)$ is the set of all vertices of $G$ that have at least one neighbour in $S$ – equivalently, it is the union of the neighbourhoods of all the vertices in $S$.

If $G$ is a bipartite graph, then a partite set $X$ of $G$ is said to satisfy Hall's condition if $|N_G(S)| \geq |S|$ for all $S \subseteq X$.

**Theorem 3.6** (Hall's Theorem)**.** *Let $(V_1, V_2)$ be a bipartition of $G$. Then $G$ has a matching saturating $V_1$ if and only if $V_1$ satisfies Hall's condition.*

*Proof.* If $G$ has a matching that saturates $V_1$, then distinct vertices $u$ and $v$ in $V_1$ have distinct neighbours via the edges of the matching incident with them. Thus, for any $S \subseteq V_1$, $|N_G(S)| \geq |S|$.

For the converse, suppose that $G$ has a maximum matching $M$ that does not saturate $V_1$. We will show that $V_1$ does not satisfy Hall's condition, by constructing a set $S$ for which $|N_G(S)| < |S|$.

Let $u$ be an $M$-unsaturated vertex in $V_1$. Let $S$ be the vertices in $V_1$ reachable by an $M$-alternating path, and let $T$ be the vertices in $V_2$ reachable by an $M$-alternating path. Note that $u \in S$.
Claim: $M$ matches $T$ with $S \setminus \{u\}$.

First, observe that any $M$-alternating path from $u$ reaches $V_2$ along edges not in $M$ and returns to $V_1$ along edges in $M$. Thus, each vertex of $S \setminus \{u\}$ is reached by an edge in $M$ from a vertex in $T$. Every vertex in $T$ must be $M$-saturated, for otherwise, $G$ would have an $M$-augmenting path, which is not possible as $M$ is a maximum matching. Therefore, any $M$-alternating path from $u$ to a vertex $y \in T$ can be extended to a vertex in $S$ via some edge of $M$. These edges define a bijection between $T$ and $S \setminus \{u\}$.

Thus, $|T| = |S|-1$, and $T \subseteq N_G(S)$. We show that $T = N_G(S)$. Suppose, to the contrary, that some vertex $y \in V_2 \setminus T$ has a neighbour $x \in S$. The edge $xy$ cannot be in $M$, as $u$ is $M$-unsaturated, and the other vertices in $S$ are matched to $T$ by $M$. Hence, an $M$-alternating path from $u$ to $x$ can be extended to $y$ by the edge $xy$, which contradicts the assumption that $y \notin T$.

Now, as $T = N_G(S)$, $|N_G(S)| = |T| = |S| - 1 < |S|$, and hence $V_1$ does not satisfy Hall's condition. □

**Corollary 3.7.** *For each positive integer $k$, every $k$-regular bipartite graph has a perfect matching.*

*Proof.* Let $(V_1, V_2)$ be a bipartition of a $k$-regular graph $G$. Double-counting its edges gives

$$k|V_1| = |E(G)| = k|V_2|,$$

so $|V_1| = |V_2|$. Consequently, any matching that saturates $V_1$ is perfect. We prove that such a matching exists by verifying Hall's condition.

Consider any $S \subseteq V_1$. Then the induced subgraph $H = \langle S \cup N_G(S) \rangle$ is a bipartite subgraph in which every vertex of $S$ has degree $k$. Hence, $k|S| = \sum_{v \in N_G(S)} \deg_H v \leq k|N_G(S)|$, which implies that $|N_G(S)| \geq |S|$. That is, $V_1$ satisfies Hall's condition, as required. □

### 3.2.2 König's theorem

For every graph, each vertex cover contains at least one end-vertex of every edge in a matching. Consequently, $\alpha_0(G) \geq \beta_1(G)$. For bipartite graphs, equality always holds.

**Theorem 3.8** (König)**.** *If $G$ is bipartite, then*

$$\alpha_0(G) = \beta_1(G).$$

*Proof.* Let $(V_1, V_2)$ be a bipartition of $G$, let $M$ be a maximum matching, and let $U$ be the set of $M$-unsaturated vertices in $V_1$. Starting at the vertices in $U$, follow $M$-alternating paths whose first edges are not in $M$. Let $Z_1$ and $Z_2$ be the sets of vertices reached in $V_1$ and $V_2$, respectively, and set

$$C = (V_1 \setminus Z_1) \cup Z_2.$$

The set $C$ is a vertex cover. Indeed, if an edge $xy$, with $x \in V_1$ and $y \in V_2$, had neither end in $C$, then $x \in Z_1$ and $y \notin Z_2$. If $xy \notin M$, the alternating search would reach $y$; if $xy \in M$, the way in which $x$ was reached would imply that $y \in Z_2$. Both cases are contradictions.

Every vertex in $Z_2$ is $M$-saturated, since otherwise the alternating path reaching it would be $M$-augmenting. Every vertex in $V_1 \setminus Z_1$ is also $M$-saturated, because all unsaturated vertices of $V_1$ belong to $U \subseteq Z_1$. Moreover, every edge of $M$ has exactly one end in $C$. It follows that $|C| = |M| = \beta_1(G)$. Since every vertex cover has at least $|M|$ vertices, $C$ is a minimum vertex cover, and hence $\alpha_0(G) = \beta_1(G)$. □

## 3.3 Applications: Assignment and Scheduling

Many assignment problems can be represented by a bipartite graph. One partite set represents applicants, residents, workers, or tasks; the other represents jobs, placements, machines, or time slots. An edge records an allowable assignment. A matching assigns distinct resources, and a

matching that saturates the first partite set assigns every applicant or task.

**Example 3.9** (Job assignment)**.** Suppose that three applicants $a, b, c$ are eligible for jobs $p, q, r$, with

$$N(a) = \{p, q\}, \qquad N(b) = \{p\}, \qquad N(c) = \{q, r\}.$$

Hall's condition holds, and the matching $\{bp, aq, cr\}$ assigns every applicant to an eligible job. If both $a$ and $b$ were eligible only for $p$, Hall's condition would fail for $\{a, b\}$, correctly identifying the obstruction.

**Example 3.10** (Task scheduling)**.** Let one partite set consist of tasks and the other of available time slots. Join a task to the slots in which it may be performed. A matching that saturates all tasks is a feasible schedule in which no two tasks use the same slot. Hall's theorem says that such a schedule exists precisely when every collection of tasks collectively has at least as many available slots as it has tasks.

**Exercise 3.3.** Give an example of a maximal matching that is not maximum, and an example of a minimal vertex cover that is not minimum.

**Exercise 3.4.** For the graph in Example 3.3, list all maximum matchings and all minimum vertex covers, and verify König's theorem directly.

**Exercise 3.5.** Four tasks must be assigned to four time slots. Their available slots are

$$N(t_1) = \{s_1, s_2\}, \qquad N(t_2) = \{s_2\},$$
$$N(t_3) = \{s_2, s_3\}, \qquad N(t_4) = \{s_3, s_4\}.$$

Use Hall's theorem to determine whether all four tasks can be scheduled, and find a feasible schedule if one exists.